

Def. Let F be a field. Given a polynomial $f(x) \in F[x]$ over F , the Extension $E \stackrel{\text{def}}{=} F(\alpha_1, \dots, \alpha_n)$ where $\alpha_1, \dots, \alpha_n \in \bar{F}$ are roots of f .

is called the splitting field of $f(x)$.

Further, the collection of polynomials $\{f_i(x) \mid i \in I\}$, the intersection of splitting fields of f_i is also called the splitting field.

Thm. Let $\varphi: F \rightarrow F'$ be an isomorphism of fields.

Given $f(x) = a_n x^n + \dots + a_1 x + a_0 \in F[x]$, let $f'(x) = \varphi(a_n) x^n + \dots + \varphi(a_1) x + \varphi(a_0) \in F'[x]$.

Let E be the splitting field for $f(x)$ over F ,

and E' be the splitting field for $f'(x)$ over F' .

Then, φ can be extended on E to E' , isomorphically.

Dummit. sec 13.4.

Prop. Let K be a finite extension of F .

K/F is splitting field

if and only if

every irreducible polynomial in $F[x]$ that has a root in K splits over K .

Proof. Suppose that K is the splitting field of $\{f_i(x) \mid i \in \{1, 2, \dots, n\}, f_i \in F[x]\}$. Since K/F is finite, the collection $\{f_i\}$ was chosen finite. Simply, we say that K is a splitting field of $f = f_1 \cdots f_n \in F[x]$. $\leftarrow$ 약 필요한 조건은 아님.

Let $p(x) \in F[x]$ be an irreducible polynomial such that $p(\alpha) = 0$ for some $\alpha \in K$.

Then, for any other root $\beta \in F$ of $p(x)$, there is a conjugate isomorphism

$\psi: F(\alpha) \rightarrow F(\beta)$ which fixes F and maps α to β .

Now, by the Extension Theorem, there is an Extension of ψ ,

$\overline{\psi}: K \rightarrow \overline{F}$ such that $\overline{\psi}|_{F(\alpha)} = \psi$. But, since K/F is splitting field and $\overline{\psi}$ fixes F , so $\overline{\psi}[K] = K$. Therefore, $F(\beta) \subseteq K$, so $\beta \in K$.

Conversely, suppose that the below. Since K/F is finite, we can write

$K = F(\alpha_1, \dots, \alpha_r)$. Now, let $f_i \in F[x]$ be a minimal polynomial of α_i .

Let L be the splitting field of $f_1, \dots, f_r$ over F . By the assumption,

all f_i split over K , so $L \subseteq K$. Conversely, each α_i is a root of f_i ,

so $K = F(\alpha_1, \dots, \alpha_r) \subseteq L$. Consequently, $K = L$. $\square$

Failed try: $\Rightarrow$

Let $f(x) \in F[x]$ be an irreducible polynomial such that for some $\alpha \in K$, $f(\alpha) = 0$.

If $\alpha \in F$, then f is just degree 1 polynomial, so there is nothing to prove.

If $\alpha \notin F$, then α is a root of for some f_3 , and since f_3 splits over K ,

so the irreducible f is also minimal polynomial of α , so f divides f_3 , therefore f splits over K .

α 가 어떤 f_3 의 근일 필요는 없다!

Ex) $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ 은 $(x^2-2)(x^2-3)$ 의 splitting field.

하지만 $\sqrt{2} + \sqrt{3}$ 의 minimal polynomial은 $x^4 - 10x^2 + 1$.

Prop. Let K_1, K_2 be finite extensions of F contained in the field K , and suppose that K_1 and K_2 both are splitting fields over F . Then, $K_1 K_2$ and $K_1 \cap K_2$ both are splitting fields over F .